Conductivity at conical crossings

Abstract

We consider independent electrons in a periodic crystal in their ground state, and turn on a uniform electric field at some prescribed time. We rigorously define the current per unit area and study the static conductivity of a two-dimensional semimetal. The conductivity tensor is isotropic and each conical intersection contributes as $\frac{1}{16}$ to the total conductivity.

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1 Introduction

We consider a two-dimensional perfect crystal with periodic lattice $\mathcal{R} \subset \mathbb{R}^2$. We use atomic units and a spinless model, with Hamiltonian

$$H = -\frac{1}{2}\Delta + V \quad \text{on } L^2(\mathbb{R}^2;\mathbb{C}), \tag{1}$$

where $V \in H^1_{\mathrm{per}}(\Omega;\mathbb{R})$ and Ω is a unit cell of $\mathcal{R}$. At zero temperature, the ground-state density matrix is given by

$$\gamma_0 = \mathbb{1}(H \leq \mu_{\mathrm{F}}), \tag{2}$$

where $\mu_{\mathrm{F}} \in \mathbb{R}$ is the Fermi level.

The purpose of this article is to analyze mathematically the behavior of the current appearing in the crystal when a uniform external field is turned on instantaneously at the initial time $t = 0$. Fix $e_\beta, e_\alpha \in \mathbb{R}^2$ once and for all, for the field and observation directions respectively. They need not be normalized, linearly independent, or orthogonal. Put $x_\beta = x \cdot e_\beta$ and $p = -i\nabla$. For $\varepsilon \geq 0$,

$$H_\varepsilon = H - \varepsilon x_\beta, \tag{3}$$

where H_ε denotes the self-adjoint closure of its restriction to $C_c^\infty(\mathbb{R}^2; \mathbb{C})$ (see Proposition 2.2). The state at $t \geq 0$ is

$$\gamma_\varepsilon(t) = e^{-itH_\varepsilon} \gamma_0 e^{itH_\varepsilon}. \tag{4}$$

The current is defined as

$$j_\varepsilon(t) = \underline{\mathrm{Tr}}(J\gamma_\varepsilon(t)), \tag{5}$$

where $\underline{\mathrm{Tr}}$ is the trace per unit area, defined in Section 2.1, and

$$J = p \cdot e_\alpha. \tag{6}$$

For time-reversible 2D semimetals such as graphene, the time-averaged conductivity

$$\sigma = \lim_{t\to\infty} \frac{1}{t} \int_0^t \lim_{\varepsilon\to 0^+} \frac{j_\varepsilon(t')}{\varepsilon}\, dt' \tag{7}$$

has a finite value equal to $\frac{1}{16} e_\alpha \cdot e_\beta$ times the number of Dirac points in the Brillouin zone.

Note that our results use an averaging in time, and we are unable to conclude anything about what would be the naive definition of the conductivity

$$\lim_{t\to\infty} \lim_{\varepsilon\to 0^+} \frac{j_\varepsilon(t)}{\varepsilon}. \tag{8}$$

A form of averaging of time fluctuations is always necessary to infer zero-frequency behavior from step responses in non-dissipative systems, even in the linear case. The easiest way to see this is by the very simple model for the relationship between an input $I(t)$ and an output $O(t)$:

$$i\dot{O}(t) = \omega O(t) + I(t). \tag{9}$$

This simplified model describes a forced oscillator with eigenfrequency ω, and arises from the linear response of the time-dependent Schrödinger equation of a two-level system. For a constant input I_0, there is a steady state solution $O_0 = \widehat{R}_0 I_0$, where $\widehat{R}_0 = -\frac{1}{\omega}$ is the zero-frequency transfer function of the system. However, since this system is oscillatory, this steady state may never be reached: if I is brutally switched on at time 0 with $I(t) = O(t) = 0$, if $t \leq 0$, $I(t) = I_0$ if $t > 0$, then $O(t) = O_0(1 - \mathrm{e}^{-i\omega t}) = \widehat{R}_0 I_0 (1 - \mathrm{e}^{-i\omega t})$ and we cannot define $\widehat{R}_0$ as the limit of $O(t)/I_0$ when t goes to infinity. However, by averaging we obtain

$$\lim_{t\to\infty} \frac{1}{t} \int_0^t \frac{O(t')}{I_0}\, dt' = \widehat{R}_0.$$

In our results in the linear response regime, we consider the quantity

$$j^{\mathrm{LR}}(t) = \lim_{\varepsilon\to 0^+} \frac{j_\varepsilon(t)}{\varepsilon}.$$

and then compute the conductivity in the limit $t \to \infty$. This order of limits is necessary to ensure that the electronic state never deviates significantly from its equilibrium. Considering the opposite limit, i.e. the infinite-time dynamics of $j_\varepsilon(t)$ at ε fixed, is an extremely hard problem, as it is for any dynamics of non-dissipative systems. Our method of proof is based on the standard gauge change $\widetilde{\psi}(x,t) = e^{-i\varepsilon t x_\beta} \psi(x,t)$ that transforms the constant in time but non-spatially-periodic Hamiltonian H_ε into the time-dependent Hamiltonian $\widetilde{H}_\varepsilon(t) = \frac{1}{2}(p + \varepsilon t e_\beta)^2 + V$. This Hamiltonian is spatially periodic, and the study of its dynamics can be reduced via Bloch–Floquet theory to that of its fibers

$H_{k+\varepsilon t e_\beta}$ acting on periodic functions (Section 3). Fiber by fiber, this time-dependent Hamiltonian can then be treated using the tools of time-dependent perturbation theory (Section 4). For each fixed t, we use linear response to expand $j_\varepsilon(t)$ to first order in ε (Proposition 4.1).

The structure of the paper is as follows. We describe our results in Section 2: we define the current in Proposition 2.2, and state Theorem 2.4. We devote Section 3 to preliminaries about the regularity and Bloch decomposition of the current. Section 4 states and proves the linear response result. Section 5 is devoted to the proof of the conductivity formula. Finally two short Appendices are devoted to technical issues.

2 Main results: electrical current in periodic materials

2.1 Notation

We denote by $\mathcal{R}^*$ the dual lattice

$$\mathcal{R}^* = \{K \in \mathbb{R}^2 : K \cdot R \in 2\pi\mathbb{Z} \text{ for all } R \in \mathcal{R}\},$$

by Ω a unit cell in physical space, and by $\mathcal{B}$ a unit cell in reciprocal space, called the Brillouin zone. We write $\partial_\alpha = e_\alpha \cdot \nabla_k$ and $\partial_\beta = e_\beta \cdot \nabla_k$ for the two fixed directional derivatives.

The *$\mathcal{R}$-periodic Lebesgue and Sobolev spaces* are denoted by

$$L^p_{\mathrm{per}} := \left\{u \in L^p_{\mathrm{loc}}(\mathbb{R}^2;\mathbb{C}) \mid u \ \mathcal{R}\text{-periodic}\right\},$$
$$H^s_{\mathrm{per}} := \left\{u \in H^s_{\mathrm{loc}}(\mathbb{R}^2;\mathbb{C}) \mid u \ \mathcal{R}\text{-periodic}\right\}.$$

The space of bounded linear operators on a Hilbert space $\mathcal{H}$ is denoted by $\mathcal{L}(\mathcal{H})$, and the Schatten class of bounded operators $A \in \mathcal{L}(\mathcal{H})$ such that $\mathrm{Tr}(|A|^p) < \infty$ by $\mathfrak{S}_p(\mathcal{H})$. For $R \in \mathcal{R}$, we denote by τ_R the translation operator formally defined by $\tau_R\phi = \phi(\cdot - R)$. Depending on the context, τ_R will be seen as a unitary operator on $L^2(\mathbb{R}^2;\mathbb{C})$, or as a linear operator on some $\mathcal{R}$-translation invariant subspace of $\mathcal{D}'(\mathbb{R}^2;\mathbb{C})$. A bounded operator on $L^2(\mathbb{R}^2;\mathbb{C})$ is called $\mathcal{R}$-periodic if it commutes with τ_R for all $R \in \mathcal{R}$. An unbounded self-adjoint operator on $L^2(\mathbb{R}^2;\mathbb{C})$ is called $\mathcal{R}$-periodic if its resolvent is $\mathcal{R}$-periodic. A bounded $\mathcal{R}$-periodic operator $A \in \mathcal{L}(L^2(\mathbb{R}^2;\mathbb{C}))$ is called locally trace-class if $\chi A \chi \in \mathfrak{S}_1(L^2(\mathbb{R}^2;\mathbb{C}))$ for any compactly supported function $\chi \in L^\infty(\mathbb{R}^2;\mathbb{C})$. For $p \geq 1$, we denote by $\mathfrak{S}_{p,\mathrm{per}}$ the space of $\mathcal{R}$-periodic operators $A \in \mathcal{L}(L^2(\mathbb{R}^2;\mathbb{C}))$ such that $|A|^p$ is locally trace class. Any operator $A \in \mathfrak{S}_{1,\mathrm{per}}$ has a density $\rho_A \in L^1_{\mathrm{per}}$ characterized by

$$\forall \chi \in C^\infty_{\mathrm{c}}(\mathbb{R}^2;\mathbb{C}), \quad \mathrm{Tr}(A\chi) = \int_{\mathbb{R}^2} \rho_A \chi.$$

The trace per unit area of an operator $A \in \mathfrak{S}_{1,\mathrm{per}}$ is defined as

$$\underline{\mathrm{Tr}}(A) = \frac{1}{|\Omega|}\,\mathrm{Tr}_{L^2(\mathbb{R}^2;\mathbb{C})}(\mathbb{1}_\Omega A \mathbb{1}_\Omega) = \fint_\Omega \rho_A,$$

where $\mathbb{1}_\Omega$ is the characteristic function of the unit cell Ω, and $\fint_\Omega$ is a shorthand notation for $\frac{1}{|\Omega|}\int_\Omega$. This formula is independent of the choice of the unit cell Ω.

Since we are dealing here with periodic materials, we will use the Bloch transform (also called Bloch-Floquet transform) [2]. For $K \in \mathcal{R}^*$, let T_K be the unitary multiplication operator on L^2_{per} defined by

$$\forall v \in L^2_{\mathrm{per}}, \quad (T_K v)(x) = \mathrm{e}^{-iK\cdot x} v(x) \quad \text{for a.a. } x \in \mathbb{R}^2,$$

and

$$L^2_{\mathrm{qp}}(L^2_{\mathrm{per}}) := \left\{\mathbb{R}^2 \ni k \mapsto u_k \in L^2_{\mathrm{per}} \mid \int_{\mathcal{B}} \|u_k\|^2_{L^2_{\mathrm{per}}}\, dk < \infty,\ u_{k+K} = T_K u_k \text{ for all } K \in \mathcal{R}^* \text{ and a.a. } k \in \mathbb{R}^2\right\},$$

the Hilbert space of $\mathcal{R}^*$-quasi-periodic $L^2_{\rm per}$-valued functions on $\mathbb{R}^2$ endowed with the inner product

$$\langle u, v\rangle_{L^2_{\rm qp}(L^2_{\rm per})} = \fint_{\mathcal{B}} \langle u_k, v_k\rangle_{L^2_{\rm per}}\, dk.$$

Here and below, the subscript qp refers to the quasi-periodicity property. The Bloch transform then is the unitary map from $L^2(\mathbb{R}^2;\mathbb{C})$ to $L^2_{\rm qp}(L^2_{\rm per})$ defined for $u \in C^\infty_{\rm c}(\mathbb{R}^2;\mathbb{C})$ by

$$\forall k \in \mathbb{R}^2, \quad \forall x \in \mathbb{R}^2, \quad u_k(x) = \sum_{R\in\mathcal{R}} u(x+R)\, {\rm e}^{-ik\cdot(x+R)}. \tag{10}$$

Its inverse is given by

$$u(x) = \fint_{\mathcal{B}} {\rm e}^{ik\cdot x} u_k(x)\, dk, \quad \text{for a.a. } x \in \mathbb{R}^2. \tag{11}$$

Any $\mathcal{R}$-periodic operator $A \in \mathcal{L}(L^2(\mathbb{R}^2;\mathbb{C}))$ is decomposed by the Bloch transform in the sense that there exists a function $k \mapsto A_k$ in $L^\infty_{\rm qp}(\mathcal{L}(L^2_{\rm per}))$ such that for any $u \in L^2(\mathbb{R}^2;\mathbb{C})$ and almost all $k \in \mathbb{R}^2$, $(Au)_k = A_k u_k$, and

$$A_{k+K} = T_K A_k T_K^*, \quad \text{for all } K \in \mathcal{R}^* \text{ and a.a. } k \in \mathbb{R}^2. \tag{12}$$

The A_k's are called the fibers of the operator A. If $A \in \mathfrak{S}_{1,\rm per}$, then the function $k \mapsto A_k$ is in $L^1_{\rm qp}(\mathfrak{S}_1(L^2_{\rm per}))$, the function $k \mapsto {\rm Tr}(A_k)$ is in $L^1_{\rm loc}(\mathbb{R}^2)$, $\mathcal{R}^*$-periodic, and we have

$$\underline{\rm Tr}(A) = (2\pi)^{-2} \int_{\mathcal{B}} {\rm Tr}(A_k)\, dk.$$

The Bloch decomposition theorem can be extended to unbounded $\mathcal{R}$-periodic self-adjoint operators using the resolvent [2].

In the case of the periodic Hamiltonian operator H given by (1), we have

$$H_k = \frac{1}{2}(-i\nabla + k)^2 + V. \tag{13}$$

For each $k \in \mathbb{R}^2$, H_k is a bounded below self-adjoint operator on $L^2_{\rm per}$ with domain $H^2_{\rm per}$ and compact resolvent. Let $(\lambda_{n,k})_{n\in\mathbb{N}^*}$ be the non-decreasing sequence of eigenvalues of H_k counting multiplicities

$$\lambda_{1,k} \le \lambda_{2,k} \le \lambda_{3,k} \le \cdots, \quad \lim_{n\to\infty} \lambda_{n,k} = +\infty,$$

and we use the convention $\lambda_{0,k} = -\infty$. We denote by $(u_{n,k})_{n\in\mathbb{N}^*} \in (H^2_{\rm per})^{\mathbb{N}^*}$ an $L^2_{\rm per}$-orthonormal basis of associated eigenfunctions:

$$H_k u_{n,k} = \lambda_{n,k} u_{n,k}, \quad \langle u_{m,k}, u_{n,k}\rangle_{L^2_{\rm per}} = \delta_{m,n}.$$

For $n \ge 1$ and $k \in \mathbb{R}^2$, set

$$P_{n,k} = \mathbb{1}(H_k \le \lambda_{n,k}), \qquad P_{0,k} = 0.$$

Whenever $\lambda_{n,k} < \lambda_{n+1,k}$, $P_{n,k}$ is the spectral projector associated with the lowest n eigenvalues, counting multiplicities:

$$P_{n,k} = \sum_{m=1}^{n} |u_{m,k}\rangle\langle u_{m,k}|. \tag{14}$$

Since H_k is quasi-periodic, so is $P_{n,k}$, and the eigenvalues $\lambda_{n,k}$ are $\mathcal{R}^*$-periodic functions of k. By a min-max argument (see e.g. [2]), there exists $\underline{C_1}, \overline{C_1} \in \mathbb{R}$, and $\underline{C_2}, \overline{C_2} > 0$ such that

$$\underline{C_1} + \underline{C_2} n \le \lambda_{n,k} \le \overline{C_1} + \overline{C_2} n. \tag{15}$$

The ground-state density matrix γ_0 has Bloch fibers

$$P_k = \mathbb{1}(H_k \le \mu_{\rm F}). \tag{16}$$

Their ranks are uniformly bounded by the eigenvalue estimate above. The current operator J is also $\mathcal{R}$-periodic, with fibers

$$J_k = (p+k)\cdot e_\alpha = \partial_\alpha H_k.$$

Lastly, for each $q \in \mathbb{R}^2$, we denote by G_q the unitary multiplication operator on $L^2(\mathbb{R}^2;\mathbb{C})$ defined by

$$\forall u \in L^2(\mathbb{R}^2;\mathbb{C}), \quad (G_q u)(x) = \mathrm{e}^{iq\cdot x}u(x) \quad \text{for a.a. } x \in \mathbb{R}^2. \tag{17}$$

The operator G_q is not $\mathcal{R}$-periodic, except when $q \in \mathcal{R}^*$ (in which case G_q is fibered, with $G_{q,k} = T_{-q}$ for all k). However, for any $\mathcal{R}$-periodic operator $A \in \mathcal{L}(L^2(\mathbb{R}^2;\mathbb{C}))$ and any $q \in \mathbb{R}^2$, the operator $G_q A G_q^*$ is $\mathcal{R}$-periodic and its Bloch decomposition is given by

$$(G_q A G_q^*)_k = A_{k-q}, \quad \text{for a.a. } k \in \mathbb{R}^2. \tag{18}$$

2.2 The Bloch theorem

Before attacking the well-posedness of the current $j_\varepsilon(t) = \underline{\mathrm{Tr}}(J\gamma_\varepsilon(t))$ for $\varepsilon, t \neq 0$, we first study an easier special case.

Proposition 2.1 (Bloch theorem)**.** *The current satisfies*

$$\begin{aligned} &j_0(t) = 0, \quad \forall t \geq 0 \quad \textit{(no current in the absence of external field)},\\ &j_\varepsilon(0) = 0, \quad \forall \varepsilon \geq 0 \quad \textit{(continuity of the current at } t = 0\textit{)}. \end{aligned}$$

Proof. We have

$$j_0(t) = j_\varepsilon(0) = \frac{1}{4\pi^2}\int_{\mathcal{B}} \mathrm{Tr}(\partial_\alpha H_k P_k)\, dk.$$

Since V is real, $H_{-k} = \overline{H_k}$ and $P_{-k} = \overline{P_k}$, where the bar denotes complex conjugation. Moreover, $\partial_\alpha H_{-k} = -\overline{\partial_\alpha H_k}$. Thus the real-valued, $\mathcal{R}^*$-periodic function $k \mapsto \mathrm{Tr}(\partial_\alpha H_k P_k)$ is odd, and its integral vanishes. □

2.3 Definition of the current

For $\varepsilon > 0$, the operator $H_\varepsilon = H - \varepsilon x_\beta$ is not $\mathcal{R}$-periodic, and we would naively expect that the density matrix $\gamma_\varepsilon(t)$ is not either. Yet, this operator is in fact $\mathcal{R}$-periodic. Physically, this is due to the fact that although $-\varepsilon x \cdot e_\beta$ is not periodic, its negative gradient is constant, hence periodic. The proof relies on the standard gauge transform

$$\widetilde{\psi}(x,t) = G_{-\varepsilon t e_\beta}\psi(x,t) = e^{-i\varepsilon t x_\beta}\psi(x,t), \tag{19}$$

where G_q was defined in (17), and the gauge-transformed operators

$$U_\varepsilon(t,t') = G_{-\varepsilon t e_\beta} e^{-i(t-t')H_\varepsilon} G^*_{-\varepsilon t' e_\beta}, \tag{20}$$

and

$$\widetilde{\gamma}_\varepsilon(t) = G_{-\varepsilon t e_\beta}\gamma_\varepsilon(t) G^*_{-\varepsilon t e_\beta} = U_\varepsilon(t)\gamma_0 U_\varepsilon(t)^*, \tag{21}$$

where $U_\varepsilon(t) = U_\varepsilon(t,0)$. Through this change of gauge, the dynamics induced by H_ε is equivalent to that induced by the time-dependent periodic Hamiltonian

$$\widetilde{H}_\varepsilon(t) = G_{-\varepsilon t e_\beta} H_\varepsilon G^*_{-\varepsilon t e_\beta} + \varepsilon x_\beta = \frac{1}{2}(p + \varepsilon t e_\beta)^2 + V. \tag{22}$$

Its Bloch fibers are

$$\widetilde{H}_{\varepsilon,k}(t) = \frac{1}{2}(p + k + \varepsilon t e_\beta)^2 + V = H_{k+\varepsilon t e_\beta}. \tag{23}$$

We sum up these arguments in the proposition below, together with elements that we shall use for defining the current.

Proposition 2.2. *Let $V \in H^1_{\mathrm{per}}(\Omega;\mathbb{R})$.*

1. *For every $\varepsilon \in \mathbb{R}$, H_ε is essentially self-adjoint on $C_c^\infty(\mathbb{R}^2;\mathbb{C})$ and therefore generates a unitary group on $L^2(\mathbb{R}^2;\mathbb{C})$.*

2. *The operator $\widetilde{H}_\varepsilon(t)$ is self-adjoint with domain $H^2(\mathbb{R}^2;\mathbb{C})$ and $\mathcal{R}$-periodic. The propagator $U_\varepsilon(t,t')$ is $\mathcal{R}$-periodic, with fibers $U_{\varepsilon,k}(t,t')$ solving*

$$i\partial_t U_{\varepsilon,k}(t,t') = H_{k+\varepsilon t e_\beta} U_{\varepsilon,k}(t,t'), \qquad U_{\varepsilon,k}(t',t') = \mathrm{Id}. \tag{24}$$

3. *For $t \geq 0$, $J\gamma_\varepsilon(t) \in \mathfrak{S}_{1,\mathrm{per}}$, and*

$$\begin{aligned} j_\varepsilon(t) &= \frac{1}{4\pi^2}\int_{\mathcal{B}} \mathrm{Tr}\left(\partial_\alpha H_{k+\varepsilon t e_\beta}\widetilde{\gamma}_{\varepsilon,k}(t)\right)\, dk \\ &= \frac{1}{4\pi^2}\int_{\mathcal{B}} \mathrm{Tr}\left(\partial_\alpha H_{k+\varepsilon t e_\beta} U_{\varepsilon,k}(t) P_k U_{\varepsilon,k}(t)^*\right)\, dk. \end{aligned} \tag{25}$$

The results of Proposition 2.2 are proved in Section 3 for completeness. The situation can be summed up in the commutative diagrams of Figure 1.

$$\begin{array}{ccc} \gamma_0 & \xrightarrow{e^{-itH_\varepsilon}} & \gamma_\varepsilon(t) \\ \downarrow{\scriptstyle \mathrm{Id}} & & \downarrow{\scriptstyle G_{-\varepsilon t e_\beta}} \\ \gamma_0 & \xrightarrow{U_\varepsilon(t)} & \widetilde{\gamma}_\varepsilon(t) \end{array} \qquad \begin{array}{ccc} \{P_k\}_k & \longrightarrow & \{\gamma_{\varepsilon,k}(t)\}_k \\ \downarrow{\scriptstyle \mathrm{Id}} & & \downarrow{\scriptstyle k\mapsto k+\varepsilon t e_\beta} \\ \{P_k\}_k & \xrightarrow{\{U_{\varepsilon,k}(t)\}_k} & \{\widetilde{\gamma}_{\varepsilon,k}(t)\}_k \end{array}$$

Figure 1: Relations between the density matrices and their Bloch fibers. An arrow labelled by a unitary U denotes conjugation by U. The right lower arrow means $\widetilde{\gamma}_{\varepsilon,k}(t) = \gamma_{\varepsilon,k+\varepsilon t e_\beta}(t)$.

2.4 Semimetals

The position of the Fermi level in the band diagram is key to determining the electronic properties of the medium. We assume the following.

Assumption 2.3 (semimetal)**.** *There are $N, M \geq 1$ and distinct points $k_1, \ldots, k_M$ modulo $\mathcal{R}^*$ such that*

$$\lambda_{N,k} < \mu_{\mathrm{F}} < \lambda_{N+1,k} \qquad (k \notin \{k_1,\ldots,k_M\} \bmod \mathcal{R}^*).$$

For each $i = 1, \ldots, M$, the two bands meet at an isolated conical crossing:

$$\lambda_{N-1,k_i} < \lambda_{N,k_i} = \mu_{\mathrm{F}} = \lambda_{N+1,k_i} < \lambda_{N+2,k_i}, \tag{26}$$

$$\lambda_{N,k} = \mu_{\mathrm{F}} - v_i|k-k_i| + O(|k-k_i|^2), \tag{27}$$

$$\lambda_{N+1,k} = \mu_{\mathrm{F}} + v_i|k-k_i| + O(|k-k_i|^2), \tag{28}$$

where $v_i > 0$.

The Fermi surface is the finite set

$$\mathcal{S} = \{k_1,\ldots,k_M\} \bmod \mathcal{R}^*. \tag{29}$$

For $k \notin \mathcal{S}$, $P_k = P_{N,k}$ has rank N. Its values at the crossings do not affect any Brillouin-zone integral. Since V is real, $H_{-k} = \overline{H_k}$: the system has time-reversal symmetry. The regularity assumption $V \in H^1_{\mathrm{per}}$ will be used to prove the Dyson expansion in Proposition 4.1.

2.5 Conductivity

In the following, $O(f(\varepsilon,t))$ denotes a quantity bounded by $Cf(\varepsilon,t)$, where C may depend on the potential, the Fermi level and the fixed directions, but not on t or ε.

Theorem 2.4. *Under Assumption 2.3, the iterated limit exists and*

$$\sigma = \lim_{T\to\infty} \frac{1}{T}\int_0^T \left(\lim_{\varepsilon\to 0^+} \frac{j_\varepsilon(t)}{\varepsilon}\right) dt = \frac{M}{16}\, e_\alpha \cdot e_\beta.$$

The value of the conductivity is universal: it does not depend on the characteristics of the Hamiltonian, but only on the number of conical crossings. The conductivity tensor is isotropic and each conical intersection contributes 1/16.

3 Bloch decomposition and regularity of the current

In this section, we prove Proposition 2.2. We first point out an alternative interpretation of the gauge change. Formally, $\gamma_\varepsilon(t)$ satisfies

$$i\partial_t \gamma_\varepsilon = [H, \gamma_\varepsilon] - \varepsilon[x_\beta, \gamma_\varepsilon].$$

The commutator $[x_\beta, \gamma_\varepsilon]$ is $\mathcal{R}$-periodic, with fibers $i\partial_\beta \gamma_{\varepsilon,k}$. Therefore the fibers satisfy

$$i\partial_t \gamma_{\varepsilon,k} + i\varepsilon\partial_\beta \gamma_{\varepsilon,k} = [H_k, \gamma_{\varepsilon,k}] = L_{H_k}\gamma_{\varepsilon,k},$$

where $L_{H_k} = [H_k, \cdot]$ is the Liouvillian associated with H_k (see Section 4.1). The left-hand side is a linear advection equation, which suggests the method of characteristics: setting

$$\widetilde{\gamma}_{\varepsilon,k}(t) = \gamma_{\varepsilon,k+\varepsilon t e_\beta}(t), \tag{30}$$

we obtain

$$i\partial_t \widetilde{\gamma}_{\varepsilon,k}(t) = [H_{k+\varepsilon t e_\beta}, \widetilde{\gamma}_{\varepsilon,k}(t)].$$

The gauge transform $G_{-\varepsilon t e_\beta}$, equivalent to the change of variable (30), makes these remarks rigorous. We now prove Proposition 2.2.

Proof of the first assertion. The essential self-adjointness of H_ε follows from an extension of the Faris-Lavine theorem [1, Theorem X.38]. Let $\mathcal{C} = C_c^\infty(\mathbb{R}^2;\mathbb{C})$ be the set of infinitely differentiable, compactly supported functions.

Lemma 3.1 (Faris-Lavine criterion)**.** *Let V and W be real-valued measurable functions on $\mathbb{R}^2$, with $W \in L^2_{\mathrm{loc}}(\mathbb{R}^2;\mathbb{R})$. Suppose that*

1. *there exist $c > 0$ and $f \geq 0$ such that $W(x) \geq -c|x|^2 - f$, for a.a. $x \in \mathbb{R}^2$;*
2. *$-\frac{1}{2}\Delta + V + W + 2c|x|^2$ is essentially self-adjoint on $\mathcal{C}$;*
3. *for some $a < 1$, $-\frac{a}{2}\Delta + V$ is bounded below on $\mathcal{C}$.*

Then $-\frac{1}{2}\Delta + V + W$ is essentially self-adjoint on $\mathcal{C}$.

The proof of the above lemma is postponed until Appendix A.1.

We apply Lemma 3.1 with $W = -\varepsilon x_\beta$ and $c = 1$, noting that

$$-\varepsilon x \cdot e_\beta \geq -|x|^2 - \frac{\varepsilon^2 |e_\beta|^2}{4}.$$

In two dimensions, $V \in L^2_{\mathrm{per}}$ is infinitesimally bounded with respect to $-\Delta$, hence also with respect to the harmonic oscillator $-\frac{1}{2}\Delta + 2|x|^2$. The linear potential $-\varepsilon x_\beta$ is likewise infinitesimally bounded with respect to this oscillator. The Kato–Rellich theorem therefore gives the essential self-adjointness

of $-\frac{1}{2}\Delta + V - \varepsilon x_\beta + 2|x|^2$ on the core $\mathcal{C}$. Also, $-\frac{a}{2}\Delta + V$ is bounded below for every $a > 0$. Lemma 3.1 gives that H_ε is essentially self-adjoint on $\mathcal{C}$, and its propagator is the unitary group $(e^{-itH_\varepsilon})_{t\in\mathbb{R}}$.

Proof of the second assertion. The self-adjointness of

$$\widetilde{H}_\varepsilon(t) = \frac{1}{2}(p + \varepsilon t e_\beta)^2 + V$$

on $H^2(\mathbb{R}^2;\mathbb{C})$ follows from the Kato–Rellich theorem. To show the well-posedness of the dynamics, it suffices to study its fibers $H_{k+\varepsilon t e_\beta}$. We use the following lemma on time-dependent perturbations of the free-particle Hamiltonian on L^2_{per}.

Lemma 3.2. *Let $H_0 := -\frac{1}{2}\Delta$ be the free-particle Hamiltonian on L^2_{per}, and a C^1 map $H_1 : [0,T] \to \mathcal{L}(H^2_{\mathrm{per}}, L^2_{\mathrm{per}})$, taking its values in the set of H_0-bounded self-adjoint operators on L^2_{per} with relative bound lower than 1, that is: there exist $0 < a < 1$ and $b > 0$ such that*

$$\forall t \in [0,T], \quad \forall \phi \in H^2_{\mathrm{per}}, \quad \|H_1(t)\phi\|_{L^2_{\mathrm{per}}} \le a\|H_0\phi\|_{L^2_{\mathrm{per}}} + b\|\phi\|_{L^2_{\mathrm{per}}}. \tag{31}$$

Then, for all $t \in [0,T]$, the operator defined by $H(t) = H_0 + H_1(t)$ is self-adjoint on L^2_{per} with domain H^2_{per}, and there exists a unique unitary propagator $(\mathcal{U}(t))_{t\in[0,T]}$ on L^2_{per} such that for $t \in [0,T]$, and $\phi_0 \in H^2_{\mathrm{per}}$, $\phi : t \mapsto \mathcal{U}(t)\phi_0$ is in $C([0,T];H^2_{\mathrm{per}}) \cap C^1([0,T];L^2_{\mathrm{per}})$, and solves the time-dependent Schrödinger equation

$$i\partial_t\phi(t) = H(t)\phi(t), \qquad \phi(0) = \phi_0.$$

The proof of the above lemma is postponed to Appendix A.2.

For $t \in [0,T]$, write $H_{k+\varepsilon t e_\beta} = H_0 + H_1(t)$, with

$$H_1(t) = (k + \varepsilon t e_\beta)\cdot p + \frac{1}{2}|k + \varepsilon t e_\beta|^2 + V.$$

The Sobolev embedding $H^2_{\mathrm{per}} \subset L^\infty_{\mathrm{per}}$ and $V \in L^2_{\mathrm{per}}$ imply the conditions of Lemma 3.2; the result follows.

Proof of the third assertion. We first compute the fibers of $\gamma_\varepsilon(t)$. Using (18), we have

$$\begin{aligned}
\gamma_{\varepsilon,k}(t) &= \left(G^*_{-\varepsilon t e_\beta} U_\varepsilon(t)\gamma_0 U_\varepsilon(t)^* G_{-\varepsilon t e_\beta}\right)_k && (32)\\
&= U_{\varepsilon,k-\varepsilon t e_\beta}(t) P_{k-\varepsilon t e_\beta} U_{\varepsilon,k-\varepsilon t e_\beta}(t)^* && (33)\\
&= \sum_{n:\,\lambda_{n,k-\varepsilon t e_\beta} \le \mu_{\mathrm{F}}} |U_{\varepsilon,k-\varepsilon t e_\beta}(t)u_{n,k-\varepsilon t e_\beta}\rangle\langle U_{\varepsilon,k-\varepsilon t e_\beta}(t)u_{n,k-\varepsilon t e_\beta}|,
\end{aligned}$$

for almost every k. Since $u_{n,k} \in H^2_{\mathrm{per}}$, we deduce that

$$(J\gamma_\varepsilon(t))_k = \partial_\alpha H_k \gamma_{\varepsilon,k}(t)$$

is trace-class and finite-rank, with a uniform bound for $k \in \mathcal{B}$. The current is therefore well-defined. Since $k \mapsto \mathrm{Tr}(\partial_\alpha H_k \gamma_{\varepsilon,k}(t))$ is $\mathcal{R}^*$-periodic,

$$\begin{aligned}
j_\varepsilon(t) &= \frac{1}{4\pi^2}\int_{\mathcal{B}} \mathrm{Tr}(\partial_\alpha H_k \gamma_{\varepsilon,k}(t))\,dk\\
&= \frac{1}{4\pi^2}\int_{\mathcal{B}} \mathrm{Tr}\big(\partial_\alpha H_{k+\varepsilon t e_\beta}\gamma_{\varepsilon,k+\varepsilon t e_\beta}(t)\big)\,dk,
\end{aligned}$$

which gives (25).

4 Linear response

4.1 The Liouvillian and its partial inverse

In order to formulate and interpret our results, it is convenient to make use of the Liouvillian formalism. Recall that if h is a bounded self-adjoint operator on a Hilbert space $\mathcal{H}_{\mathrm{f}}$, the Liouvillian L_h associated with h is the bounded linear operator on $\mathcal{L}(\mathcal{H}_{\mathrm{f}})$ (such a mathematical object is sometimes called a superoperator in the physics literature) defined by

$$\forall A \in \mathcal{L}(\mathcal{H}_{\mathrm{f}}), \quad L_h A = [h, A]. \tag{34}$$

Note that if A is self-adjoint, then $L_h A$ is anti-self-adjoint ($iL_hA = i[h, A]$ is self-adjoint). The restriction of L_h to the space $\mathfrak{S}_2(\mathcal{H}_{\mathrm{f}})$ of Hilbert-Schmidt operators on $\mathcal{H}_{\mathrm{f}}$ is self-adjoint: for all $A, B \in \mathfrak{S}_2(\mathcal{H}_{\mathrm{f}})$,

$$(L_h A, B)_{\mathfrak{S}_2} = \mathrm{Tr}\,([h, A]^* B) = \mathrm{Tr}\,((A^* h - hA^*)B) = \mathrm{Tr}\,(A^*(hB - Bh)) = (A, L_h B)_{\mathfrak{S}_2}. \tag{35}$$

The operator L_h is to density matrices what the Hamiltonian h is to pure states: it is the infinitesimal generator of the norm-continuous unitary group $(\mathfrak{U}_h(t))_{t\in\mathbb{R}}$ on $\mathcal{L}(\mathcal{H}_{\mathrm{f}})$ defined by

$$\forall A \in \mathcal{L}(\mathcal{H}_{\mathrm{f}}), \quad \mathfrak{U}_h(t) A = e^{-ith} A e^{ith}. \tag{36}$$

In the case when h is an unbounded self-adjoint operator, (34) does not make sense for all $A \in \mathcal{L}(\mathcal{H}_{\mathrm{f}})$, but it is still possible to define the Liouvillian L_h as the self-adjoint generator of the strongly-continuous unitary group $(\mathfrak{U}_h(t))_{t\in\mathbb{R}}$ on $\mathfrak{S}_2(\mathcal{H}_{\mathrm{f}})$ defined by (36). The conjugation formula also acts on bounded operators in the strong operator topology.

If $\mathcal{H}_f$ is of finite-dimension N_f, the action of L_h is easily understood in an orthonormal eigenbasis $(e_n)_{1\le n\le N_f}$ of h with associated eigenvalues $\lambda_1 \le \cdots \le \lambda_{N_f}$. Then,

$$L_h |e_n\rangle\langle e_m| = (\lambda_n - \lambda_m)|e_n\rangle\langle e_m|.$$

The operator L_h is not invertible (for instance, $L_h|e_n\rangle\langle e_n| = 0$). However, it is invertible when restricted to the subspace of block off-diagonal matrices, i.e. matrices A such that $A_{nn'} = A_{mm'} = 0$ for $n, n' \le N < m, m'$ for a given N such that $\lambda_{N+1} > \lambda_N$. Its partial inverse $L^+_{h,N}$ is given by

$$L^+_{h,N}\Big(\sum_{1\le n\le N,\, N<m\le N_f} A_{mn}|e_m\rangle\langle e_n| + A_{nm}|e_n\rangle\langle e_m|\Big) = \Big(\sum_{1\le n\le N,\, N<m\le N_f} \frac{A_{mn}|e_m\rangle\langle e_n| - A_{nm}|e_n\rangle\langle e_m|}{\lambda_m - \lambda_n}\Big) \tag{37}$$

and $L^+_{h,N}$ is bounded in operator norm by $\frac{1}{\lambda_{N+1}-\lambda_N}$.

More generally, if h is an unbounded self-adjoint operator, let I be a closed bounded interval of $\mathbb{R}$, and assume that

$$g := \min\left(1, \mathrm{dist}\left(I, \sigma(h) \setminus (\sigma(h) \cap I)\right)\right) > 0.$$

The associated spectral projector is

$$P_{h,I} := \mathbb{1}_I(h) = \frac{1}{2\pi i} \oint_{\mathcal{C}} (z - h)^{-1}\, dz, \tag{38}$$

where $\mathcal{C}$ is a Cauchy contour in the complex plane such that $\sigma(h) \cap I$ is inside $\mathcal{C}$ and $\sigma(h) \setminus (\sigma(h) \cap I)$ is outside $\mathcal{C}$. Generalizing the terminology of the finite-dimensional case, we call off-diagonal operators (with respect to the splitting of $\mathcal{H}_f$ induced by $P_{h,I}$) the elements of the closed subspace

$$\mathcal{L}^{\mathrm{OD}}_{h,I} := \{A \in \mathcal{L}(\mathcal{H}_{\mathrm{f}}) \mid P_{h,I} A P_{h,I} = (1 - P_{h,I})A(1 - P_{h,I}) = 0\}$$

of $\mathcal{L}(\mathcal{H}_\mathrm{f})$. This defines a $\mathfrak{S}_2$-orthogonal splitting of operators into their diagonal and off-diagonal parts. It is easily seen that $\mathcal{L}^{\mathrm{OD}}_{h,I}$ is L_h-stable, and that L_h is invertible on $\mathcal{L}^{\mathrm{OD}}_{h,I}$ with a bounded inverse. We denote its partial inverse by $L^+_{h,I}$, extended to all of $\mathcal{L}(\mathcal{H}_f)$ by imposing that it vanishes on diagonal operators. We then have

$$L_{h,I}L^+_{h,I}A = L^+_{h,I}L_{h,I}A = P_{h,I}A(1-P_{h,I}) + (1-P_{h,I})AP_{h,I}$$

for all $A \in \mathcal{L}(\mathcal{H}_f)$.

It is easy to check that $L^+_{h,I}$ has an explicit contour integral representation:

$$L^+_{h,I}A = \frac{1}{2\pi i}\oint_{\mathcal{C}}(z-h)^{-1}[P_{h,I},A](z-h)^{-1}\,dz, \quad \forall A \in \mathcal{L}(\mathcal{H}_\mathrm{f}), \tag{39}$$

where $\mathcal{C}$ is a contour as above. From (39), we see that, when $P_{h,I}$ is of finite rank $\mathcal{N}$, $L^+_{h,I}A$ is of rank of most $2\mathcal{N}$.

4.2 Spectral derivatives and propagation bounds

For a differentiable family $H(s)$ with a common domain and a locally isolated spectral group, the contour formula (38) gives a differentiable spectral projector $P(s)$. Then

$$\dot{P}(s) = L^+_{H(s),I}[P(s),\dot{H}(s)]. \tag{40}$$

Differentiating $L_{H(s)}P(s) = 0$, we get

$$L_{H(s)}\dot{P}(s) = [P(s),\dot{H}(s)].$$

Since both $[P(s),\dot{H}(s)]$ and $\dot{P}(s)$ are off-diagonal operators (the first by direct calculation, the second by differentiating the relationship $P(s)^2 = P(s)$), we deduce (40).

We also need propagation bounds that do not require a spectral gap. Let $(H(s))_{s\in[0,T)}$ be a family with common domain D, with $H(s) \geq 1$, and set $\alpha_1 = \sup_s \|\dot{H}(s)H(s)^{-1}\| < \infty$. Let $U^\varepsilon(t,t')$ solve

$$i\partial_t U^\varepsilon(t,t') = H(\varepsilon t)U^\varepsilon(t,t'), \qquad U^\varepsilon(t',t') = \mathrm{Id}. \tag{41}$$

Then

$$\|H(\varepsilon t)U^\varepsilon(t,t')H(\varepsilon t')^{-1}\| \leq e^{\alpha_1\varepsilon(t-t')}, \tag{42}$$

$$\|H(\varepsilon t)^{1/2}U^\varepsilon(t,t')H(\varepsilon t')^{-1/2}\| \leq e^{\alpha_1\varepsilon(t-t')/2}. \tag{43}$$

Proof. Let us now prove (42). Let $\psi \in D$. For all $t \in [t', \varepsilon^{-1}T)$, we set $\psi_\varepsilon(t) = U^\varepsilon(t,t')\psi$ and $\phi_\varepsilon(t) = H(\varepsilon t)\psi_\varepsilon(t)$. We have

$$i\frac{d\phi_\varepsilon}{dt}(t) = i\frac{d}{dt}\left(H(\varepsilon t)\psi_\varepsilon(t)\right) = H(\varepsilon t)\phi_\varepsilon(t) + i\varepsilon\dot{H}(\varepsilon t)H(\varepsilon t)^{-1}\phi_\varepsilon(t),$$

from which we obtain

$$\phi_\varepsilon(t) = U^\varepsilon(t,t')H(\varepsilon t')\psi + \varepsilon\int_{t'}^t U^\varepsilon(t,t'')\dot{H}(\varepsilon t'')H(\varepsilon t'')^{-1}\phi_\varepsilon(t'')\,dt'',$$

and finally

$$\|\phi_\varepsilon(t)\|_{\mathcal{H}_\mathrm{f}} \leq \|H(\varepsilon t')\psi\|_{\mathcal{H}_\mathrm{f}} + \alpha_1\varepsilon\int_{t'}^t \|\phi_\varepsilon(t'')\|_{\mathcal{H}_\mathrm{f}}\,dt''.$$

By the Grönwall lemma,

$$\|H(\varepsilon t)U^\varepsilon(t,t')\psi\|_{\mathcal{H}_\mathrm{f}} = \|\phi_\varepsilon(t)\|_{\mathcal{H}_\mathrm{f}} \leq \|H(\varepsilon t')\psi\|_{\mathcal{H}_\mathrm{f}}e^{\alpha_1\varepsilon(t-t')}.$$

Applying this inequality to $\psi = H(\varepsilon t')^{-1}\phi$ for all $\phi \in \mathcal{H}_\mathrm{f}$ gives (42). We obtain (43) by interpolation (see e.g. [1, Section IX.4, Proposition 9]). □

For $H(s) = H_{k+se_\beta} + \mu$ we choose $\mu = 1 - \min \sigma(H)$, so that $H(s) \geq 1$, and write

$$\eta = \sup_{k \in \mathbb{R}^2} \|\partial_\beta H_k (H_k + \mu)^{-1}\| < \infty.$$

Since the occupied projectors have uniformly bounded rank and energy, the preceding estimate also gives

$$\left|\mathrm{Tr}\left(\partial_\alpha H_{k+\varepsilon t e_\beta} U_{\varepsilon,k}(t) P_k U_{\varepsilon,k}(t)^*\right)\right| \leq C e^{\eta \varepsilon t}. \tag{44}$$

For $k \notin \mathcal{S}$, we abbreviate $L_k = L_{H_k}$ and $L_k^+ = L^+_{H_k,[\lambda_{1,k},\lambda_{N,k}]}$.

4.3 Linear response

We now aim at obtaining an expansion of the current to first order in ε for a given t, based on a Dyson expansion. This is a classical computation in response theory, sometimes known as the Kubo formula. This gives a remainder that does not depend on a gap, and will therefore be useful for the study of semimetals near Dirac points.

Proposition 4.1. *Let H be the periodic Hamiltonian defined in (1). With $V \in H^1_{\mathrm{per}}$, there exists a constant $C \in \mathbb{R}_+$ such that for all $k \in \mathbb{R}^2$ such that $\lambda_{N+1,k} - \lambda_{N,k} > 0$, we have for all $\varepsilon, t \in \mathbb{R}_+$,*

$$\begin{aligned}\mathrm{Tr}\left(\partial_\alpha H_{k+\varepsilon e_\beta t} U_{\varepsilon,k}(t)\, P_k\, U_{\varepsilon,k}(t)^*\right) =& \partial_\alpha \mathrm{Tr}\left(H_k P_k\right) + \varepsilon t \partial_\alpha \partial_\beta \left(\mathrm{Tr}(H_k P_k)\right) \\ &- i\varepsilon\, \mathrm{Tr}\big(\partial_\alpha H_k (e^{-itL_k} - 1) L_k^+ \partial_\beta P_k\big) + \rho_k^\varepsilon(t),\end{aligned} \tag{45}$$

with, when $\varepsilon t \leq 1$,

$$|\rho_k^\varepsilon(t)| \leq C \varepsilon^2 t^3 (1 + t^3). \tag{46}$$

Proof. Let k be such that $\lambda_{N+1,k} - \lambda_{N,k} > 0$. Since $k' \mapsto \mathrm{Tr}\left(H_{k'} P_{k'}\right)$ is real-analytic in a neighborhood of k, we have by Hellmann-Feynman theorem

$$\partial_\alpha \mathrm{Tr}\left(H_k P_k\right) = \mathrm{Tr}\left(\partial_\alpha H_k P_k\right) \quad \text{and} \quad \partial_\alpha \partial_\beta \mathrm{Tr}\left(H_k P_k\right) = \mathrm{Tr}\left(\partial_\alpha \partial_\beta H_k P_k\right) + \mathrm{Tr}\left(\partial_\alpha H_k \partial_\beta P_k\right).$$

We also have $\partial_\alpha H_{k+\varepsilon e_\beta t} = \partial_\alpha H_k + \varepsilon t e_\alpha \cdot e_\beta$. It follows that

$$\begin{aligned}\rho_k^\varepsilon(t) =& \mathrm{Tr}\left(\partial_\alpha H_k U_{\varepsilon,k}(t)\, P_k\, U_{\varepsilon,k}(t)^*\right) - \mathrm{Tr}\left(\partial_\alpha H_k P_k\right) - \varepsilon t \mathrm{Tr}\left(\partial_\alpha H_k \partial_\beta P_k\right) \\ &+ i\varepsilon\, \mathrm{Tr}\big(\partial_\alpha H_k (e^{-itL_k} - 1) L_k^+ \partial_\beta P_k\big).\end{aligned}$$

We now expand the first term in the right-hand side of this equation. We set $\mu := 1 - \min \sigma(H)$,

$$H(s) := H_{k+se_\beta} + \mu, \quad A = \partial_\alpha H_k, \quad I_0 = [0, \frac{1}{2}(\lambda_{N,k} + \lambda_{N+1,k}) + \mu], \quad P(s) = \mathbb{1}_{I_0}(H(s)).$$

It holds

$$H(s) = h_0 + s h_1 + \frac{s^2 |e_\beta|^2}{2}$$

with $h_0 = H_k + \mu$ and $h_1 = \partial_\beta H_k$. The operators h_0, h_1 and A are self-adjoint on L^2_{per} and we have $h_0 \geq 1$ and $h_1 h_0^{-1/2}$ and $A h_0^{-1/2}$ bounded. For s in a neighborhood of zero, $P(s) = P_{k+se_\beta}$, so that $\dot{P}(0) = \partial_\beta P_k$. Let $(U^\varepsilon(t,t'))_{t,t' \in \mathbb{R}}$ be the propagator associated with the family $(H(\varepsilon t))_{t \in \mathbb{R}}$ and $U^\varepsilon(t) := U^\varepsilon(t,0)$. We have $U^\varepsilon(t,t') = e^{-i\mu(t-t')} U_{\varepsilon,k}(t,t')$ and $U^\varepsilon(t) = e^{-i\mu t} U_{\varepsilon,k}(t)$. With these notations, we have

$$\rho_k^\varepsilon(t) = \mathrm{Tr}\left(A U^\varepsilon(t)\, P(0) U^\varepsilon(t)^*\right) - \mathrm{Tr}\left(AP(0)\right) - \varepsilon t \mathrm{Tr}\left(A\dot{P}(0)\right) + i\varepsilon\, \mathrm{Tr}\Big(A(e^{-itL_0} - 1) L_0^+ \dot{P}(0)\Big),$$

where $L_0 = L_{h_0,I_0}$ and $L_0^+ = L^+_{h_0,I_0}$, and we focus on expanding the operator $U^\varepsilon(t)\, P(0) U^\varepsilon(t)^*$ close to $t = 0$.

Lemma 4.2. *We have*

$$U^\varepsilon(t)\,P(0)U^\varepsilon(t)^* = P(0) + \varepsilon\left(t\dot{P}(0) - i\left(e^{-itL_0}-1\right)\left(L_0^+\dot{P}(0)\right)\right) + \Pi_2^\varepsilon(t) \tag{47}$$

with $\rho_k^\varepsilon(t) = \mathrm{Tr}\left(A\Pi_2^\varepsilon(t)\right)$. *Moreover, when* $\varepsilon t \le 1$, *we have the bound* (46)

$$|\rho_k^\varepsilon(t)| \le C\varepsilon^2 t^3(1+t^3).$$

Lemma 4.2 closes the proof of Proposition 4.1. □

Proof of Lemma 4.2. We deduce from the Dyson expansion that

$$U^\varepsilon(t) = U^0(t) + V^\varepsilon(t) + W^\varepsilon(t),$$

where $c = |e_\beta|^2$, $U^0(t) = e^{-ith_0}$ and

$$V^\varepsilon(t) = -i\varepsilon\int_0^t U^0(t-t')t'h_1U^0(t')\,dt',$$
$$W^\varepsilon(t) = \varepsilon^2\left(-i\frac{ct^3}{6}U^0(t) - \int_0^t\int_0^{t'} U^\varepsilon(t,t')t'(h_1+\varepsilon ct'/2)U^0(t'-t'')t''(h_1+\varepsilon ct''/2)U^0(t'')\,dt''\,dt'\right).$$

This induces $U^\varepsilon(t)\,P(0)U^\varepsilon(t)^* = P(0) + \Pi_1^\varepsilon(t) + \Pi_2^\varepsilon(t)$ where

$$\Pi_1^\varepsilon(t) = V^\varepsilon(t)P(0)U^0(t)^* + \text{h.c.} = -i\varepsilon\int_0^t t'U^0(t-t')[h_1,P(0)]U^0(t-t')^*\,dt',$$
$$\Pi_2^\varepsilon(t) = V^\varepsilon(t)P(0)V^\varepsilon(t)^* + \left(W^\varepsilon(t)P(0)(U^0(t)+V^\varepsilon(t))^* + \text{h.c.}\right) + W^\varepsilon(t)P(0)W^\varepsilon(t)^*.$$

We first analyze $\Pi_1^\varepsilon(t)$ by computing

$$\begin{aligned} U^0(t-t')[h_1,P(0)]U^0(t-t')^* &= -e^{-i(t-t')L_0}L_0\dot{P}(0)\\ &= i\frac{d}{dt'}e^{-i(t-t')L_0}\dot{P}(0)\\ &= \frac{d^2}{dt'^2}e^{-i(t-t')L_0}L_0^+\dot{P}(0), \end{aligned}$$

where we have used $\dot{P}(0) = L_0^+[P(0),h_1]$ and $\dot{P}(0) = L_0^+L_0\dot{P}(0)$. Using integration by parts, we obtain

$$\Pi_1^\varepsilon(t) = \varepsilon\left(t\dot{P}(0) - i\left(e^{-itL_0}-1\right)\left(L_0^+\dot{P}(0)\right)\right)$$

and (47) follows.

We now work on the bound (46). For that purpose, we introduce the following quantities, which are independent of k, ε and t:

$$\begin{aligned} \nu_0 &= \max_{|e|\le\max(|e_\alpha|,|e_\beta|)} \|(p\cdot e)(H+\mu)^{-1/2}\|_{\mathcal{L}(L^2(\mathbb{R}^2;\mathbb{C}))},\\ \nu_1 &= \max_{|e|\le\max(|e_\alpha|,|e_\beta|)} \|(H+\mu)^{1/2}(p\cdot e)(H+\mu)^{-1}\|_{\mathcal{L}(L^2(\mathbb{R}^2;\mathbb{C}))},\\ \nu_2 &= \max_{|e|\le\max(|e_\alpha|,|e_\beta|)} \|(H+\mu)(p\cdot e)(H+\mu)^{-2}\|_{\mathcal{L}(L^2(\mathbb{R}^2;\mathbb{C}))},\\ \lambda &= \max_{k\in\mathbb{R}^2,|k-k'|\le|e_\beta|} \|(H_k+\mu)^{1/2}(H_{k'}+\mu)^{-1/2}\|_{\mathcal{L}(L^2_{\mathrm{per}})}. \end{aligned}$$

Note that $V \in H^1_{\mathrm{per}}$ ensures that the quantities ν_0, ν_1 and λ are finite. Writing $p_\beta = p\cdot e_\beta$, since $\|h_0h_1h_0^{-2}\| \le \|(H+\mu)p_\beta(H+\mu)^{-2}\|$ and

$$(H+\mu)p_\beta(H+\mu)^{-2} = p_\beta(H+\mu)^{-1} + i(e_\beta\cdot\nabla V)(H+\mu)^{-2}.$$

we deduce from $V \in H^1_{\rm per}$ that $\|(H+\mu)p_\beta(H+\mu)^{-2}\| < \infty$, hence that $\nu_2 < \infty$.

We now aim at controlling $\rho_k(t)$ thanks to ν_0, ν_1, ν_2 and λ. Using the relations $P(0) = P(0)^2$ and $P(0) = h_0^{-m} h_0^m P(0)$ with

$$\|h_0^m P(0)\| \le (\mu_{\rm F} + \mu)^m, \quad \|h_0^{-1}\| \le \|h_0^{-1/2}\| \le 1 \quad \text{and} \quad \mathrm{Rank}(P(0)) = N,$$

we deduce

$$\begin{aligned}
|\rho_k^\varepsilon(t)| &\le N\Big((\mu_{\rm F}+\mu)^2 \|AV^\varepsilon(t)h_0^{-1}\| \, \|V^\varepsilon(t)h_0^{-1}\| + 2(\mu_{\rm F}+\mu)^3 \|AW^\varepsilon(t)h_0^{-2}\| \, (\|h_0^{-1}\| + \|V^\varepsilon(t)h_0^{-1}\|) \\
&\qquad + (\mu_{\rm F}+\mu)^3 \|AW^\varepsilon(t)h_0^{-2}\| \|W^\varepsilon(t)h_0^{-1}\| \Big) \\
&\le N\nu_0 \Big((\mu_{\rm F}+\mu)^2 \|h_0^{1/2}V^\varepsilon(t)h_0^{-1}\|^2 + (\mu_{\rm F}+\mu)^3 \|h_0^{1/2}W^\varepsilon(t)h_0^{-2}\| (2 + 2\|V^\varepsilon(t)h_0^{-1}\| \\
&\qquad + \|W^\varepsilon(t)h_0^{-1}\|) \Big).
\end{aligned}$$

Next, we get

$$\begin{aligned}
\|V^\varepsilon(t)h_0^{-1}\| &\le \frac{\varepsilon t^2}{2}\|h_1 h_0^{-1}\| \le \frac{\varepsilon t^2}{2}\nu_0, \quad \|h_0^{1/2}V^\varepsilon(t)h_0^{-1}\| \le \frac{\varepsilon t^2}{2}\nu_1, \\
\|W^\varepsilon(t)h_0^{-1}\| &\le \varepsilon^2 t^3 \left(\frac{c}{6} + \nu_0\nu_1 \frac{t}{8} + (\varepsilon t) tc \left(\frac{\nu_0}{30} + \frac{\nu_1}{20} \right) + (\varepsilon t)^2 t \frac{c^2}{72} \right), \\
\|h_0^{1/2}W^\varepsilon(t)h_0^{-2}\| &\le \varepsilon^2 t^3 \lambda e^{\eta\varepsilon t/2} \left(\frac{c}{6} + \nu_1\nu_2 \frac{t}{8} + (\varepsilon t) tc \left(\frac{\nu_1}{30} + \frac{\nu_2}{20} \right) + (\varepsilon t)^2 t \frac{c^2}{72} \right).
\end{aligned}$$

It follows that there exists a constant C depending only on V, $\mu_{\rm F}$ and the fixed directions, such that

$$|\rho_k^\varepsilon(t)| \le C\varepsilon^2 t^3 \left(t + e^{\eta\varepsilon t/2} \left(1 + t(1 + (\varepsilon t)^2) + (\varepsilon t) t^2 (1 + (\varepsilon t)^3) + (\varepsilon t)^4 t^3 (1 + (\varepsilon t)^2) \right) \right),$$

which leads to (46) when $\varepsilon t \le 1$. □

5 Proof of the conductivity formula

We prove here Theorem 2.4. We decompose the integral defining $j_\varepsilon(t)$ into several parts depending whether one integrates far from the Dirac points or not.

Throughout the proof we assume $\varepsilon t \le 1$, as is sufficient for the limit $\varepsilon \to 0^+$ at fixed t. We introduce a small parameter $\delta > 0$ controlling the size of the neighborhood of the Dirac points, which is independent of t, ε. We decompose $\mathcal{B}$ as the disjoint union

$$\mathcal{B} = \mathcal{B}^\delta_{\rm out} \cup (\cup_{i=1}^M \mathcal{B}^\delta_i)$$

with

$$\mathcal{B}^\delta_i = B(k_i, \delta),$$

where $\delta > 0$ is small enough so that

$$\mathcal{B}^\delta_{\rm out} \subset \{k \in \mathcal{B},\ \lambda_{N,k} \le \mu_F - c\delta\}$$

for some constant $c > 0$. Note that this decomposition is time-reversal symmetric in the sense that

$$-\mathcal{B}^\delta_{\rm out} = \mathcal{B}^\delta_{\rm out} \quad \text{and} \quad -(\cup_{i=1}^M \mathcal{B}^\delta_i) = (\cup_{i=1}^M \mathcal{B}^\delta_i).$$

We first let $\varepsilon \to 0^+$ with t and δ fixed, then let $t \to \infty$, and finally let $\delta \to 0$. Write $\mathcal{B}^\delta_{\rm in} = \cup_{i=1}^M \mathcal{B}^\delta_i$. On $\partial\mathcal{B}^\delta_i$, the vector surface element ds points into the ball (out of $\mathcal{B}^\delta_{\rm out}$).

In the following analysis, we first treat the regions $\mathcal{B}^\delta_{\text{out}}$, where the linear response formula has a non-zero gap larger than a constant times δ. In the sets $\mathcal{B}^\delta_i$, where the gap closes, we study the structure of the Taylor expansion of the Hamiltonian H_k close to the Dirac points and construct two-band reduced Hamiltonians $H^R_{i,k}$. Then, we use the linear response Proposition 4.1, reducing successively from the Hamiltonian H to the reduced Hamiltonian $H^R_{i,k}$, and finally to the Dirac Hamiltonian

$$H^D_k = v \begin{pmatrix} 0 & k_1 - ik_2 \\ k_1 + ik_2 & 0 \end{pmatrix}, \qquad v > 0,$$

for which we can explicitly compute the current. Adding the contributions, we will obtain

$$\sigma := \lim_{t\to\infty} \lim_{\varepsilon\to 0^+} \frac{1}{\varepsilon t} \int_0^t j_\varepsilon(t')dt' = \frac{M}{16} e_\alpha \cdot e_\beta + O(\delta)$$

Finally, we will pass to the limit $\delta \to 0$.

5.1 Far from the Dirac points

We set

$$j^{\text{out}}_\varepsilon(t) := \frac{1}{4\pi^2} \int_{\mathcal{B}^\delta_{\text{out}}} \operatorname{Tr}\left(\partial_\alpha H_{k+\varepsilon t e_\beta} U_{\varepsilon,k}(t) P_k U_{\varepsilon,k}(t)^*\right) dk.$$

Using the linear response Proposition 4.1, we have

$$\begin{aligned}\operatorname{Tr}\left(\partial_\alpha H_{k+\varepsilon t e_\beta} U_{\varepsilon,k}(t) P_k U_{\varepsilon,k}(t)^*\right) &= \partial_\alpha \operatorname{Tr}(H_k P_k) + \varepsilon t \partial_\alpha \partial_\beta \operatorname{Tr}(H_k P_k) \\ &\quad - i\varepsilon \operatorname{Tr}\left(\partial_\alpha H_k (e^{-itL_k} - 1) L_k^+ \partial_\beta P_k\right) + O(\varepsilon^2 t^3 (1+t^3)).\end{aligned}$$

We treat each term separately.

- By time-reversal symmetry, the first term vanishes when integrated on $\mathcal{B}^\delta_{\text{out}}$.
- Using Stokes formula for the second term, we get

$$\int_{\mathcal{B}^\delta_{\text{out}}} \varepsilon t \partial_\alpha \partial_\beta \operatorname{Tr}(H_k P_k)\, dk = \varepsilon t \sum_{n\le N} \int_{\partial \mathcal{B}^\delta_{\text{in}}} \partial_\alpha \lambda_{n,k} (ds \cdot e_\beta). \tag{48}$$

- For the third term, the constant part vanishes when integrated on the time-reversal symmetric set $\mathcal{B}^\delta_{\text{out}}$. Indeed, complex conjugation relates k and $-k$; the trace $\operatorname{Tr}\left(\partial_\alpha H_k L_k^+ \partial_\beta P_k\right)$ is purely imaginary and odd. For the oscillatory part we use

$$\frac{1}{t} \int_0^t e^{-i\omega t'}\, dt' = \frac{e^{-i\omega t} - 1}{-i\omega t}.$$

  In the eigenbasis representation (37), only pairs $n \le N < m$ occur. On $\mathcal{B}^\delta_{\text{out}}$ their energy differences are bounded away from zero for each fixed δ. The finite number of occupied states, the uniform bounds on $\partial_\alpha H_k u_{n,k}$ and $\partial_\beta H_k u_{n,k}$, and Cauchy–Schwarz give absolute summability of the products of matrix elements. Thus the time average of the oscillatory part is $O_\delta(\varepsilon/t)$.

We are left with

$$\begin{aligned}\frac{1}{t} \int_0^t j^{\text{out}}_\varepsilon(t')\, dt' &= \frac{\varepsilon t}{8\pi^2} \sum_{n\le N} \int_{\partial \mathcal{B}^\delta_{\text{in}}} \partial_\alpha \lambda_{n,k} (ds \cdot e_\beta) \\ &\quad + O_\delta(\varepsilon/t) + O(\varepsilon^2 t^3 (1+t^3)).\end{aligned} \tag{49}$$

Here O_δ permits a constant depending on the fixed δ.

5.2 Close to Dirac points: reduction to the 2-band case

We set

$$j_\varepsilon^{\mathrm{in}}(t) := \frac{1}{4\pi^2}\sum_{i=1}^{M}\int_{\mathcal{B}_i^\delta} \mathrm{Tr}\left(\partial_\alpha H_{k+\varepsilon t e_\beta} U_{\varepsilon,k}(t)\, P_k\, U_{\varepsilon,k}(t)^*\right)\, dk. \tag{50}$$

Using the linear response Proposition 4.1, we have for almost all $k \in \mathcal{B}_{\mathrm{in}}^\delta$ and all $\varepsilon, t \geq 0$ with $\varepsilon t \leq 1$,

$$\begin{aligned}\mathrm{Tr}\left(\partial_\alpha H_{k+\varepsilon e_\beta t} U_{\varepsilon,k}(t)\, P_k\, U_{\varepsilon,k}(t)^*\right) =& \partial_\alpha \mathrm{Tr}\left(H_k P_k\right) + \varepsilon t \partial_\alpha\partial_\beta\left(\mathrm{Tr}(H_k P_k)\right)\\ &- i\varepsilon\,\mathrm{Tr}\big(\partial_\alpha H_k (e^{-itL_k}-1) L_k^+ \partial_\beta P_k\big) + O(\varepsilon^2 t^3(1+t^3)).\end{aligned}$$

Using (44), the equality $\mathrm{Tr}\left(H_k P_k\right) = \sum_{n=1}^N \lambda_{n,k}$ and Assumption 2.3, it is easily seen that the left-hand side, as well as the first and fourth terms of the right-hand side of that equation, are bounded uniformly in k and therefore integrable on $\mathcal{B}_i^\delta$. Besides, for $k \in \mathcal{B}_{\mathrm{in}}^\delta$, the second term is bounded by a constant multiple of $(1/|k-k_i|)$ as $k \to k_i$, and is therefore integrable. It follows that the third term is also integrable on $\mathcal{B}_i^\delta$.

We treat the three leading terms of the right-hand side in sequence.

- The first term vanishes when integrated on the time-reversal symmetric set $\cup_{i=1}^M \mathcal{B}_i^\delta$.
- For the second, using Stokes formula, we get

$$\int_{\mathcal{B}_i^\delta} \varepsilon t \partial_\alpha\partial_\beta\left(\mathrm{Tr}(H_k P_k)\right) dk = -\varepsilon t \sum_{n\leq N}\int_{\partial\mathcal{B}_i^\delta} \partial_\alpha \lambda_{n,k}(ds\cdot e_\beta),$$

  so that the corresponding term in (50) cancels the contribution (49) from $\mathcal{B}_{\mathrm{out}}^\delta$.
- For the third term, we use

$$\begin{aligned}&\mathrm{Tr}\big(\partial_\alpha H_k (e^{-itL_k}-1) L_k^+ \partial_\beta P_k\big) = \mathrm{Tr}\big(\partial_\alpha H_k (e^{-itL_k}-1)(L_k^+)^2 [P_k, \partial_\beta H_k]\big)\\ &= \sum_{n\leq N}\sum_{m>N} (e^{-it(\lambda_{n,k}-\lambda_{m,k})}-1)\frac{\langle u_{n,k}, \partial_\beta H_k u_{m,k}\rangle\langle u_{m,k}, \partial_\alpha H_k u_{n,k}\rangle}{(\lambda_{m,k}-\lambda_{n,k})^2} - \text{c.c.}\end{aligned}$$

  with the sum converging from the asymptotics (15).

  When $n \neq N$ or $m \neq N+1$, the denominators in that equation are bounded from below independently of δ. The constant term vanishes when integrated over the time-reversal symmetric set $\cup_{i=1}^M \mathcal{B}_i^\delta$, and the oscillatory term can be treated using the formula

$$\frac{1}{t}\int_0^t e^{-i\omega t'} dt' = \frac{e^{-i\omega t}-1}{-i\omega t}$$

  with $\omega = \lambda_{n,k} - \lambda_{m,k}$ bounded away from zero independently of δ.

Putting all the results of the previous two sections together, we get that

$$\begin{aligned}\frac{1}{t}\int_0^t j_\varepsilon(t')dt' &= \frac{1}{t}\int_0^t j_\varepsilon^{\mathrm{out}}(t')dt' + \frac{1}{t}\int_0^t j_\varepsilon^{\mathrm{in}}(t')dt'\\ &= -\frac{i\varepsilon}{4\pi^2 t}\Bigg(\sum_{i=1}^M \int_0^t dt' \int_{\mathcal{B}_i^\delta} (e^{-it'(\lambda_{N,k}-\lambda_{N+1,k})}-1)\\ &\qquad\qquad \times \frac{\langle u_{N,k}, \partial_\beta H_k u_{N+1,k}\rangle\langle u_{N+1,k}, \partial_\alpha H_k u_{N,k}\rangle}{(\lambda_{N+1,k}-\lambda_{N,k})^2} dk - \text{c.c.}\Bigg)\\ &\quad + O_\delta(\varepsilon t^{-1}) + O(\varepsilon^2 t^3(1+t^3)).\end{aligned}$$

At this stage of the proof, only two modes are involved in the formula giving the current, namely the two modes that cross at the Fermi level. Everything happens as for a two-band model that we now study. We write for short

$$\frac{1}{t}\int_0^t \frac{j_\varepsilon(t')}{\varepsilon}\,dt' = -\frac{i}{4\pi^2 t}\sum_{i=1}^{M}\int_0^t I_i^R(\delta,t')\,dt' + O_\delta(t^{-1}) + O(\varepsilon t^3(1+t^3)),$$

with

$$I_i^R(\delta,t) = \int_{\mathcal{B}_i^\delta}(e^{-it(\lambda_{N,k}-\lambda_{N+1,k})}-1)\frac{\langle u_{N,k},\partial_\beta H_k u_{N+1,k}\rangle\langle u_{N+1,k},\partial_\alpha H_k u_{N,k}\rangle}{(\lambda_{N+1,k}-\lambda_{N,k})^2}\,dk - \text{c.c.}.$$

5.3 Close to the Dirac points: the local model

We now are interested in the computation of $I_i^R(\delta,t)$. In the following, we drop the index i and assume without loss of generality that $k_i = 0$. We write $v = v_i$.

Hypothesis (26) implies that, for k small enough, the Bloch Hamiltonian H_k has exactly two eigenvalues (counting multiplicities) close to $\mu_{\rm F}$. Consider an arbitrary orthonormal basis (v_0, w_0) of $\mathrm{Ran}(P_{N+1,0} - P_{N-1,0})$. For all k small enough, we can construct an orthonormal basis (v_k, w_k) of $\mathrm{Ran}(P_{N+1,k} - P_{N-1,k})$ by Löwdin orthonormalization of $((P_{N+1,k} - P_{N-1,k})v_0, (P_{N+1,k} - P_{N-1,k})w_0)$, and set

$$H_k^R = [v_k|w_k]^* H_k [v_k|w_k] = \begin{pmatrix} \langle v_k, H_k v_k\rangle & \langle v_k, H_k w_k\rangle \\ \langle w_k, H_k v_k\rangle & \langle w_k, H_k w_k\rangle \end{pmatrix}.$$

It follows that the reduced Hamiltonian

$$H_k\Big|_{\mathrm{Ran}(P_{N+1,k}-P_{N-1,k})}$$

is equivalent through a unitary transform that depends analytically on k to the reduced 2×2 Hamiltonian

$$H_k^R = \sum_{p=0}^{3} b^p(k)\sigma_p, \tag{51}$$

where

$$\sigma_0 = \mathrm{Id}_{\mathbb{C}^2},\quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},\quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are the Pauli matrices, and $(b^p)_{p=0,1,2,3}$ are real-valued analytic functions of k in a neighborhood of 0.

The matrix H_k^R has eigenvalues

$$\lambda_\pm(k) = b^0(k) \pm \sqrt{\sum_{p=1}^{3} b^p(k)^2}.$$

It follows that

$$b^0(k) = \mu_F + O(|k|^2),\quad b^p(k) = v\langle q^p, k\rangle + O(|k|^2),\quad p = 1,2,3,$$

where the $(q^p)_{p=1,2,3}$ are the rows of a 3×2 matrix Q with orthonormal columns, so that

$$H_k^R = \mu_F + v(Qk)\cdot\sigma + O(|k|^2). \tag{52}$$

Let $R \in \mathrm{SO}(3)$ be a rotation matrix that maps $\mathrm{Ran}(Q)$ to $\mathrm{Span}(e_1^0, e_2^0)$, where (e_1^0, e_2^0, e_3^0) is the canonical basis of $\mathbb{R}^3$. Write R as a rotation through an angle θ about a unit vector n. The unitary matrix

$$U = \cos(\theta/2)\mathrm{Id}_2 - i\sin(\theta/2)n\cdot\sigma$$

satisfies, by the Pauli matrix identities,

$$R_{pq} = \frac{1}{2}\operatorname{Tr}(\sigma_p U \sigma_q U^*).$$

It follows that

$$\begin{aligned}\operatorname{Tr}\big(\sigma_3 U H_k^R U^*\big) &= v\sum_{q=1}^{3}(Qk)_q \operatorname{Tr}(\sigma_3 U\sigma_q U^*) + O(|k|^2)\\ &= 2v\langle e_3^0, RQk\rangle + O(|k|^2)\\ &= O(|k|^2).\end{aligned}$$

Up to a unitary transform, we can therefore assume Q to be a 2×2 matrix in (52).

5.4 The two-band case: reduction to the Dirac Hamiltonian

Reduction to H_k^R For $k\neq 0$, let $\lambda_{\pm,k}^R$ be the larger and smaller eigenvalues of H_k^R respectively, and $u_{\pm,k}^R$ associated orthonormal eigenvectors in $\mathbb{C}^2$. We have $\lambda_{-,k}^R = \lambda_{N,k}, \lambda_{+,k}^R = \lambda_{N+1,k}$, and

$$[v_k|w_k]u_{+,k}^R = e^{i\theta_+(k)}u_{N+1,k}, \qquad [v_k|w_k]u_{-,k}^R = e^{i\theta_-(k)}u_{N,k}$$

for some phases $\theta_\pm(k)\in\mathbb{R}$. We have

$$\begin{aligned}\partial_\alpha H_k^R &= [v_k|w_k]^*\partial_\alpha H_k[v_k|w_k] + \partial_\alpha[v_k|w_k]^*[v_k|w_k]H_k^R + H_k^R[v_k|w_k]^*\partial_\alpha[v_k|w_k]\\ &= [v_k|w_k]^*\partial_\alpha H_k[v_k|w_k] + O(|k|)\end{aligned}$$

where we have used for the first line that H_k commutes with $P_{N+1,k}-P_{N-1,k} = [v_k|w_k][v_k|w_k]^*$, and for the second that $H_k^R = \mu_F\mathrm{Id}_2 + O(|k|)$ and $\partial_\alpha\Big([v_k|w_k]^*[v_k|w_k]\Big) = \partial_\alpha\mathrm{Id}_2 = 0$. We therefore obtain

$$\begin{aligned}\langle u_{+,k}^R, \partial_\alpha H_k^R u_{-,k}^R\rangle &= e^{-i(\theta_+(k)-\theta_-(k))}\langle u_{N+1,k}, \partial_\alpha H_k u_{N,k}\rangle + O(|k|)\\ \langle u_{-,k}^R, \partial_\beta H_k^R u_{+,k}^R\rangle &= e^{+i(\theta_+(k)-\theta_-(k))}\langle u_{N,k}, \partial_\beta H_k u_{N+1,k}\rangle + O(|k|).\end{aligned}$$

Since $(\lambda_{+,k}^R - \lambda_{-,k}^R)$ is bounded from below by a constant multiple of $|k|$, it follows that

$$I^R(\delta,t) = \int_{B(0,\delta)}(e^{-it(\lambda_{-,k}^R-\lambda_{+,k}^R)}-1)\frac{\langle u_{-,k}^R, \partial_\beta H_k^R u_{+,k}^R\rangle\langle u_{+,k}^R, \partial_\alpha H_k^R u_{-,k}^R\rangle}{(\lambda_{+,k}^R-\lambda_{-,k}^R)^2}dk - \text{c.c.} + O(\delta). \tag{53}$$

Reduction to H_k^D By standard results of perturbation theory [2, Chapter XII] applied to $H_k^R = \mu_F + v(Qk)\cdot\sigma + O(|k|^2)$ with gap greater than a constant multiple of $|k|$,

$$\lambda_{\pm,k}^R = \mu_F + \lambda_{\pm,k}^Q + O(|k|^2) \text{ and } u_{\pm,k}^R = u_{\pm,k}^Q + O(|k|)$$

where the superscript Q refers to eigenvalues and appropriately chosen orthonormal eigenvectors of the Hamiltonian

$$H_k^Q = v(Qk)\cdot\sigma.$$

It follows that

$$I^R(\delta,t) = I^Q(\delta,t) + O(\delta),$$

where $I^Q(\delta,t)$ is defined similarly to (53) as

$$I^Q(\delta,t) = \int_{B(0,\delta)}(e^{-it(\lambda_{-,k}^Q-\lambda_{+,k}^Q)}-1)\frac{\langle u_{-,k}^Q, \partial_\beta H_k^Q u_{+,k}^Q\rangle\langle u_{+,k}^Q, \partial_\alpha H_k^Q u_{-,k}^Q\rangle}{(\lambda_{+,k}^Q-\lambda_{-,k}^Q)^2}dk - \text{c.c.}.$$

We perform the change of variable $k' = Qk$ (recall that Q is orthogonal) and obtain

$$I^Q(\delta,t) = (Qe_\beta)^T I^D(\delta,t)(Qe_\alpha),$$

where the coefficients $I^D_{ij}(\delta,t)$ of the 2×2 matrix $I^D(\delta,t)$ are given by

$$I^D_{ij}(\delta,t) = \int_{B(0,\delta)} (e^{-it(\lambda^D_{-,k}-\lambda^D_{+,k})}-1)\frac{\langle u^D_{-,k},\partial_{k_j}H^D_k u^D_{+,k}\rangle\langle u^D_{+,k},\partial_{k_i}H^D_k u^D_{-,k}\rangle}{(\lambda^D_{+,k}-\lambda^D_{-,k})^2}dk - \text{c.c.}$$

and the superscript D refers to the Dirac Hamiltonian

$$H^D_k = vk\cdot\sigma.$$

The Dirac Hamiltonian H^D_k We finish by computing $I^D(\delta,t)$ explicitly. Let $k = r(\cos\theta,\sin\theta)$. We have

$$\lambda^D_{\pm,k} = \pm vr,\quad u^D_{+,k} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ e^{i\theta}\end{pmatrix}\quad\text{and}\quad u^D_{-,k} = \frac{1}{\sqrt{2}}\begin{pmatrix}-e^{-i\theta}\\ 1\end{pmatrix}.$$

By an explicit calculation, we obtain

$$\int_0^{2\pi}(e^{-it(\lambda^D_{-,k}-\lambda^D_{+,k})}-1)\frac{\langle u^D_{-,k},\partial_{k_i}H^D_k u^D_{+,k}\rangle\langle u^D_{+,k},\partial_{k_j}H^D_k u^D_{-,k}\rangle}{(\lambda^D_{+,k}-\lambda^D_{-,k})^2}\,d\theta - \text{c.c.} = i\pi\frac{1}{2r^2}\sin(2vrt)\delta_{ij}.$$

It follows that

$$\begin{aligned}\frac{1}{t}\int_0^t I^D_{ij}(\delta,t')\,dt' &= \frac{i\pi}{2}\delta_{ij}\frac{1}{t}\int_0^t\int_0^\delta \frac{\sin(2vrt')}{r}\,dr\,dt' = \frac{i\pi}{4t}\delta_{ij}\int_0^\delta\frac{1-\cos(2vrt)}{vr^2}\,dr\\ &= \frac{i\pi}{4}\delta_{ij}\int_0^{\delta vt}\frac{1-\cos(2r')}{(r')^2}\,dr' = \frac{i\pi^2}{4}\delta_{ij} + O\left((\delta t)^{-1}\right).\end{aligned}$$

We finally get by summing all the estimates

$$\frac{1}{t}\int_0^t\frac{j_\varepsilon(t')}{\varepsilon}\,dt' = \frac{M}{16}e_\alpha\cdot e_\beta + O(\delta) + O_\delta(t^{-1}) + O(\varepsilon t^3(1+t^3)),$$

hence the result, since Q is orthogonal. The uniform remainder in Proposition 4.1 also permits the limit $\varepsilon\to 0^+$ to pass under each integral over a fixed bounded time interval.

A Proofs of two technical lemmata

A.1 Proof of Lemma 3.1

Proof. We use the proof of the Faris–Lavine theorem in [1]. It consists in verifying the following two hypotheses of [1, Theorem X.37]. Let $A = -\frac{1}{2}\Delta + W + V$ and $K = A + 2c|x|^2 + b$, where $b\in\mathbb{R}$ will be specified below:

$$\textit{there exists } h\textit{, such that for any } \phi\in\mathcal{C},\quad \|A\phi\|\le h\|K\phi\|; \tag{54}$$

$$\textit{for some } \ell\textit{, for any } \phi\in\mathcal{C},\quad |(A\phi,K\phi)-(K\phi,A\phi)|\le \ell\|K^{\frac{1}{2}}\phi\|^2. \tag{55}$$

By hypothesis *3* in Lemma 3.1 and the conditions on W, it is possible to choose b so that $K\ge 1$. As quadratic forms on $\mathcal{C}$,

$$K^2 = (A+b)^2 + 4c\sum_{j=1}^2 x_j(A+b+c|x|^2)x_j - 4c.$$

Hypotheses *1* and *3* guarantee that $A+b+c|x|^2$ is bounded below. Hence, increasing the value of b if necessary to make this operator positive, we have

$$\|(A+b)\phi\|^2_{L^2}\le\|K\phi\|^2_{L^2} + 8c\|\phi\|^2_{L^2},$$

which proves (54).

For (55), we observe that

$$\pm i[A,K] = \pm 2c(x\cdot(-i\nabla)+(-i\nabla)\cdot x) \le 2c\left((-\Delta)+|x|^2\right) \le \ell K,$$

where we have used

$$(-\Delta)+|x|^2 \pm (x\cdot(-i\nabla)+(-i\nabla)\cdot x) = (-i\nabla \pm x)^2 \ge 0$$

and

$$K = \left(\frac{a}{2}(-\Delta)+V\right) + (W+c|x|^2) + \frac{1-a}{2}(-\Delta) + c|x|^2 + b \ge e((-\Delta)+|x|^2),$$

where $e = \min(c, \frac{1-a}{2}) > 0$ and where b is chosen so that

$$b - f + \min\,\sigma\left(\frac{a}{2}(-\Delta)+V\right) \ge 0.$$

This proves (55). Hence A is essentially self-adjoint on $\mathcal{C}$. □

A.2 Proof of Lemma 3.2

Proof. By the Kato-Rellich theorem, for any $0 \le t \le T$, $H(t)$ is self-adjoint on $L^2_{\rm per}$ with domain $H^2_{\rm per}$, and bounded below. We will show that there exists $\mu > 0$ so that the graph norm of $(H(t)+\mu)$ for any $0 \le t \le T$ is equivalent to the $H^2_{\rm per}$-norm. Together with the assumed C^1 dependence on t, this verifies the hypotheses of the propagator theorem [1, Theorem X.70], and proves Lemma 3.2.

We have for any $\mu > 0$, $0 \le t \le T$ and $\phi \in H^2_{\rm per}$,

$$\|(H(t)+\mu)\phi\|_{L^2_{\rm per}} \le (1+a)\|H_0\phi\|_{L^2_{\rm per}} + (b+\mu)\|\phi\|_{L^2_{\rm per}} \le (1+a+b+\mu)\|\phi\|_{H^2_{\rm per}},$$

and so the graph norm is controlled by the $H^2_{\rm per}$-norm.

For the other inequality, we relate the resolvent of $H(t)$ to that of H_0 by a bounded operator, with bounded inverse. Notice that, for any $\mu > 0$, since H_0 is positive,

$$\forall\ 0 \le t \le T, \quad (H(t)+\mu) = (1 + H_1(t)(H_0+\mu)^{-1})(H_0+\mu).$$

Furthermore,

$$\forall\ 0 \le t \le T, \quad \|H_1(t)(H_0+\mu)^{-1}\| \le a\|H_0(H_0+\mu)^{-1}\| + b\|(H_0+\mu)^{-1}\| \le a + \frac{b}{\mu}.$$

and so, for $\mu > \frac{b}{1-a}$, the operator $1 + H_1(t)(H_0+\mu)^{-1}$ is bounded and invertible with bounded inverse in $L^2_{\rm per}$. Therefore $(H(t)+\mu)^{-1}$ is bounded from $L^2_{\rm per}$ to $H^2_{\rm per}$, which means there exists $C > 0$ such that, for any $\phi \in H^2_{\rm per}$ and $0 \le t \le T$,

$$\|\phi\|_{H^2_{\rm per}} = \|(H(t)+\mu)^{-1}(H(t)+\mu)\phi\|_{H^2_{\rm per}} \le C\|(H(t)+\mu)\phi\|_{L^2_{\rm per}},$$

which concludes the proof. □

References

[1] Michael C. Reed and Barry Simon. *Methods of Modern Mathematical Physics. II. Fourier analysis, self-adjointness.* Academic Press, New York, 1975.

[2] Michael C. Reed and Barry Simon. *Methods of Modern Mathematical Physics. IV. Analysis of operators.* Academic Press, New York, 1978.